

AXONA

Machine-Oriented Grounding for AI-Built Apps

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The Axona AI Grounding v4.48.0 2026-07-27

Axona AI Grounding — kernel 4.48.0

This file is the complete, self-contained grounding for an AI system building an application on the Axona protocol. It matches the network it targets: **kernel 4.48.0 / wire 4.0, deployed on testnet (wss://testnet.axona.net)**. Everything below is exact and current; nothing outside this file is required. If this version does not match the bridge you are connecting to, request the matching grounding file.

Division of authority. You will usually build from an application design document *plus* this file. The design document is the authority on **what** to build; this file is the sole authority on **how every protocol call is shaped**. When a design document names a protocol operation in prose (“issues a kill for that message”, “subscribes to the feed”), that prose never licenses a guessed signature — the call shape comes from here, exactly. The most expensive bug yet observed in a generated Axona app came from inferring an API’s arguments from design-doc prose.

Axona is a serverless peer-to-peer pub/sub network. Every participant is a peer. Topics are addresses derived from descriptors; messages route to an emergent per-topic root and fan out through a distribution tree. There is no central server, message broker, or database.

HARD RULES — violations cause silent failure or thrown errors

1. **A topic is a descriptor object, never a string.** { region?, owner?, name, write? }. Publisher and subscriber must use IDENTICAL fields — the topic ID is a pure hash of them. A mismatch (one names region, the other omits it) means two different topics and **no delivery, with no error**.
2. **Every publish names its signer.** `peer.pub(topic, msg, { signWith })` — `signWith` is an author identity or the ANONYMOUS sentinel. Omitting it throws PUBLISH_NO_PUBLISH_IDENTITY. There is no default signer.
3. **Two identities, never interchangeable.** The *node* identity (`createNodeIdentity`) is the connection — it has a location and forms the node ID; it NEVER signs content. The *author* identity (`createAuthorIdentity`) signs content — it has NO location and NO node ID.
4. **await peer.ready() before first pub/sub.** Publishing on a cold mesh is survivable (the kernel retries) but unpredictable. `ready()` resolves when the routing mesh is warm.
5. **A publish returns no delivery acknowledgment — ever.** This is a privacy invariant (author and transport identities must stay uncorrelated). To confirm delivery, subscribe to the topic yourself and watch for your own `msgId` to arrive.
6. **Message payloads: 15 KB is the reliable cross-browser floor.** Hard ceiling 256 KB, but anything over 15 KB may fail on real WebRTC channels. For larger payloads use `std/chunk` (below), never a bigger single publish.
7. **Messages expire.** Default hold ~24 h (48 h ceiling). Re-publishing the identical payload from the same author yields the SAME `msgId` and refreshes the hold (an upsert; subscribers are not re-notified). Axona is a live messaging fabric, not permanent storage.
8. **kill is the only retraction** and its signature is `peer.kill(topic, msgId, { signWith })` — the **descriptor comes first**, exactly like `pub`. `peer.kill(msgId)` (omitting the topic) is the most

common mistake observed in the field and throws. The kill must be signed by the same author key that signed the original, and is best-effort (receivers already holding the plaintext keep it). Anonymous messages cannot be killed. `unpub` and `touch` do not exist in 4.x — do not call them.

9. **Persist the author key if user identity should survive restarts.** `createAuthorIdentity({ persistAs: 'myapp:author' })` (browser `localStorage`). Everything else (node identity, subscriptions) can be ephemeral and re-created each run.
10. **Node 20+ or a browser over HTTPS.** The kernel needs Web Crypto (`crypto.subtle`) and ES modules. No build step, no bundler required.
11. **An ephemeral publisher exits via `await peer.leave()`, never a bare `process.exit()`.** A short-lived process (bot, script, one-shot job) that publishes and dies immediately can take its message with it. Since 4.30.0 `leave()`'s drain holds until the published history has left the node — `await peer.pub(...); await peer.leave();` is the durable pattern. For a critical one-shot post, additionally confirm from an INDEPENDENT fresh session (`pull(msgId)` or `sub since:'all'`).

Install

`npm install github:axona-net/axona-protocol#v4.48.0`

`package.json` must contain `"type": "module"`.

```
import {
  connect, // the one-call bootstrap (also at
  ↪ @axona/protocol/connect.js)
  AxonaPeer, AxonaDomain, NeuronNode, ANONYMOUS,
  createNodeIdentity, createAuthorIdentity,
  deriveTopicId, metricTopic, KERNEL_VERSION,
} from '@axona/protocol';
import { webTransport } from '@axona/protocol/transport/web/index.js';
import { makeMessage, readMessage } from '@axona/protocol/std/message.js';
import { publishChunkedBytes, receiveChunkedBytes } from '@axona/protocol/std/chunk.js';
```

Canonical peer bootstrap (copy exactly)

```
const { peer, author, status, disconnect } = await connect({
  bridge: 'wss://testnet.axona.net',
  location: { lat: 38.0, lng: -77.0 }, // the user's real location
  author: 'myapp:author', // string = durable (persistAs); true = ephemeral;
  ↪ false = none
});
// status: { ready, peers, ms, reason } — never rejects; ready:false on timeout

Teardown: await disconnect();
```

`connect()` is the ONLY bootstrap an application needs — every case is an option, not a reason to assemble by hand: `k` (routing set size, default 20), `ready` (forwarded to `peer.ready()`; `false` skips the

wait), author (string=durable / true=ephemeral / false=none), transport (custom/sim), nodeId-identity (durable/pre-minted), domain (share one mesh across several peers in a process), persist (persistence adapter), rootReplicas / maxPublishBytes / synaptomeMaintain (advanced tuning), web (extra webTransport options).

Driving the lifecycle yourself (sim / tests only — rare)

The constructors + `peer.start/join/integrate/ready` are ADVANCED building blocks that `connect()` composes in the correct order. Do NOT hand-assemble a peer for an application — it is how apps silently skipped self-integration and self-rooted their topics as **singletons**, losing cross-region delivery. If you genuinely drive the lifecycle (a sim harness, a test), you MUST call `peer.integrate()` after start, or the peer sits at the passive-adoption churn floor:

```
const peer = new AxonaPeer({ domain: new AxonaDomain({ k: 20 }),
                           node, nodeIdIdentity, transport });
await transport.start(nodeIdIdentity.id);
await peer.start();
await peer.ready();
await peer.integrate(); // REQUIRED by hand – connect() does this for you
```

Topics

A topic descriptor: { region?, owner?, name, write? }

Field	Type	Meaning
region	'useast' '0x89' 137	Geographic cell the topic lives in. All three forms are equivalent. Omitted → defaults to the publisher's own region (co-located peers converge; cross-region readers must then know that region). Prefer explicit.
owner	64-hex Author ID absent	Namespaces the topic under an author.
name	non-empty string	The human label. Required.
write	'open' 'owner'	Who may publish. Defaults: no owner → 'open'; owner present → 'owner'.

The three canonical shapes:

```

const lobby = { region: 'useast', name: 'lobby' }; // open room: anyone
↳ reads/writes
const feed = { region: 'useast', owner: me.authorId, name: 'posts' }; // owner-only feed
↳ (write defaults 'owner')
const inbox = { region: 'useast', owner: me.authorId, name: 'inbox', write: 'open' }; //
↳ anyone writes TO me

```

- `await deriveTopicId(descriptor)` → 66-hex topic ID (a shareable read handle). `sub/pull/metrics` accept a descriptor **or** this ID; `pub` always needs the full descriptor.
- Publishing to someone else's `write: 'owner'` topic throws / is rejected at the network with `WRITE_POLICY_VIOLATION`.

Publish / subscribe

```

// Publish (signed). Returns the msgId (64-hex content hash). NO delivery ack.
const msgId = await peer.pub(topic, message, { signWith: author });
// message: any JSON-serializable value. Anonymous: { signWith: ANONYMOUS }.

// Subscribe. handler receives an envelope for EVERY delivery, exactly once per msgId.
const sub = await peer.sub(topic, (env) => {
  if (env.deleted) { /* a retraction: drop env.msgId locally */ return; }
  // env = { msgId, seq, ts, topic, message, signerPubkey?, signature? }
  // env.signerPubkey is the author's ID (absent when anonymous)
}, { since: 'all' });

// since options: omit = live only; 'latest' = newest cached + live;
//                  'all' = full cached history + live; <unix-ms number> = newer than.

await sub.stop(); // stop this handle
await peer.unsub(topic); // stop ALL local subs for the topic + network unsubscribe

// Retract (author-only, best-effort):
await peer.kill(topic, msgId, { signWith: author });
// Subscribers receive { deleted: true, msgId, topic } on their normal handler.

```

Delivery semantics an app can rely on: - Exactly-once per `msgId` to each subscriber handler. - Publishing again with the identical payload + author = same `msgId` = refresh, not a duplicate delivery. - Late subscribers with `since: 'all'` receive cached history (bounded queue, ~24 h) then the live tail. - The kernel retries a publish internally until it confirms placement; a freshly-joined publisher gets an extra rapid burst. Your code does not retry.

Reads without subscribing

```
const env = await peer.pull(msgId, { topic });           // exact message by content hash, or
↳ null
const last = await peer.pull(null, { topic });           // topic's newest (may be slightly
↳ stale – served by the nearest replica ON PURPOSE)
if (env) use(env.message);                               // pull resolves the FULL ENVELOPE –
↳ the body is env.message
```

pull resolves the same envelope shape sub delivers ({ msgId, ts, seq, topic, message, signer-Pubkey?, signature? }) or null. **Migration note:** kernels ≤ 4.28 resolved the bare body — code that uses the pull result directly as the payload predates 4.29 and must read `env.message`. If the strict newest matters, subscribe instead of polling `pull(null)`. A successful pull extends the message's hold a little.

Topic activity (metrics)

```
const m = await peer.metrics(topic); // one-shot: latest published snapshot
// Live: subscribe to the derived metric topic
const mt = metricTopic(await deriveTopicId(topic)); // -> a topic descriptor
await peer.sub(mt, (env) => {
  const snap = JSON.parse(env.message); // { topic, ts, subscribers, current_count, seq, ...
  ↳ }
}, { since: 'latest' });
```

TIMING RULES (metrics are published ON DEMAND — the subscribe itself turns them on; there is no always-on publisher):

- First snapshot: at ROUTING LATENCY (~0.3 s typical; allow a few seconds under churn) — the root answers the moment the lease arms, independent of other users' publish/watch activity. Cadence: every ~20 s while subscribed. Publishing stops ~70 s after the last metric subscriber leaves.
- `peer.metrics()` (one-shot) normally succeeds within its 1.5 s default window; `stale: true` means the answer didn't arrive in time (churn) — retry or prefer the standing `sub()` form.
- Render silence as UNKNOWN, never as zero: a `current_count: 0` snapshot is the real “no activity” answer. Keep the subscription open in tests.

Metrics are advisory (unauthenticated counts) — never a security input.

Direct messages (peer-to-peer, not pub/sub)

```
const reply = await peer.send(targetNodeIdHex, { any: 'json' }); // RPC: awaits remote
↳ handler's return
peer.notify(targetNodeIdHex, { any: 'json' }); // fire-and-forget
peer.onMessage(async (fromNodeIdHex, message) => { return { ok: true }; });
```

Target must be a currently-known peer (`peer.peers()` lists node IDs). Direct messages address a *node* (device/session), not an author.

Presence / introspection

```
peer.peers()           // -> ['<66-hex nodeId>', ...] currently in the routing table
peer.onPeerJoin(id => {}); peer.onPeerLeave(id => {});
peer.health()          // { nodeId, synaptomeSize, subscriptions, hosting, wireVersion,
  ↪ ... }
peer.onLog('warn', (evt, data) => {}); peer.onError(err => {});
peer.onUpgradeRequired(() => { /* show "please update" */ });
```

Large payloads (files, images) — std/chunk

```
// Sender: splits into signed sub-15KB chunks, publishes each.
await publishChunkedBytes(peer, bytes /* Uint8Array */, {
  topic, signWith: author,
  meta: { filename: 'photo.jpg', mime: 'image/jpeg' },
});
// Receiver: reassembles complete payloads.
await receiveChunkedBytes(peer, topic, {
  onComplete: ({ bytes, meta }) => { /* whole file */ },
  onProgress: ({ received, total }) => {},
});
```

Interoperable message bodies — std/message

If other Axona apps may read your topic, wrap text bodies:

```
await peer.pub(topic, makeMessage('hello', { customField: 1 }), { signWith: author });
const { text } = readMessage(env.message); // tolerant reader
```

Serving without consuming — host

```
await peer.host(); // host this node's keyspace neighborhood (relay mode)
await peer.host(topic); // host one topic: store + serve it without subscribing
await peer.unhost(topic);
```

Use in long-running Node processes to keep topic history alive; browser apps rarely need it.

Agent legibility (recommended for AI-operated authors)

```
await peer.setAuthorClass('agent', { signWith: author, label: 'my-bot' });
const cls = await peer.getAuthorClass(someAuthorId); // { class:'agent'|'human'|'unstated',
  ↪ ... }
```

An AI publishing as an autonomous agent SHOULD declare 'agent'. It is voluntary provenance; absence reads as 'unstated', never 'human'.

Envelope shape (what handlers receive)

```
{  
  msgId:      "<64-hex sha256(canonical({publisher, message}))>",  
  seq:        1716638400123,          // per-author monotonic  
  ts:         1716638400000,          // author's clock (ms)  
  topic:      { region, owner, name, write }, // the SIGNED descriptor  
  message:    <your payload>,  
  signerPubkey: "<64-hex author id>", // absent if anonymous  
  signature:   "ed25519:<128-hex>"    // absent if anonymous  
}
```

Signatures are verified by the kernel at network ingress AND on delivery; `env.signerPubkey` present means verified. The signature binds the whole descriptor, so a message cannot be replayed onto a different topic.

Limits

Limit	Value
Publish size (hard)	256 KB
Publish size (reliable floor — use std/chunk above this)	15 KB
Message hold	~24 h default, 48 h ceiling; re-publish refreshes
Per-topic cached history	bounded queue (late joiners see recent history, not everything ever)
Per-publisher share of an open topic's queue	quota-bounded (one author cannot flood)
Live-publish freshness window	signed <code>ts</code> within ± 5 min of the root's clock

Error codes (err.code on thrown `AxonaError`)

Code	Thrown by	Meaning / fix
PUBLISH_NO_PUBLISH_IDENTITY	pub	Missing <code>signWith</code> . Pass an author or <code>ANONYMOUS</code> .
WRITE_POLICY_VIOLATION	pub	Signer $\neq$ owner on a <code>write: 'owner'</code> topic.
TOPIC_REGION_REQUIRED	topic derivation	Open topic with no derivable region. Name a region.
PUBLISH_PAYLOAD_TOO_LARGE	pub	Over 256 KB. Chunk it.
PUBLISH_INVALID_TOPIC / SUBSCRIBE_INVALID_TOPIC	pub/sub	Malformed descriptor (e.g. empty name).
KILL_SIGN_FAILED	kill	<code>signWith</code> is not the original author key.

Code	Thrown by	Meaning / fix
UPGRADE_REQUIRED (WS close 4426)	connect	Kernel/wire mismatch with the bridge. Install the tag matching this file.
TRANSPORT_PEER_UNREACHABLE	send	Target node not connected. Check <code>peer.peers()</code> .

`pull` returning `null` and `kill` resolving `{ ok:false }` are normal outcomes (missing/expired message; nothing to retract) — not errors.

Field-observed mistakes — every one of these has shipped in a real app

Check generated code against ALL of these before calling it done.

1. **`peer.kill(msgId)` — topic omitted.** Every content operation takes the **descriptor first**: `kill(topic, msgId, { signWith })`, exactly like `pub(topic, ...)` / `sub(topic, ...)`. There is no id-only form of anything.
2. **Reconnecting the peer to switch users/personas.** WRONG: tearing down the connection when the active author changes (it churns the mesh and drops live WebRTC links mid-ICE). One peer serves ANY number of authors — authorship is chosen per call via `{ signWith }`. Reconnect only when the bridge URL changes.
3. **Sharing a topic without its full descriptor.** An invite, directory ad, QR code, or link that transmits only name (or name + region) sends the recipient to a DIFFERENT topic whenever the original has an owner — `{region, owner, name, write}` ALL fold into the topic id (HARD RULE 1). Always transmit the complete descriptor.
4. **Deriving presence/recency from arrival time.** A `since:'all'` (or `'latest'`) subscription replays history: a message that arrives NOW may have been published hours ago. Recency logic (online lists, “last seen”) must read `env.ts` (publish time), never the local clock at delivery — and must keep the LATEST publish time per author (never let an older replayed record overwrite a fresher one; replay order is not guaranteed). For ephemeral signals like heartbeats, subscribe with `since` omitted (live-only) — replaying stale heartbeats is pure noise.
5. **Faking encryption with public inputs.** The protocol signs messages; it does NOT encrypt them, and it provides NO encrypt-to-author primitive. An author ID is **public** (it is `env.signerPubkey` on every signed message) — any scheme that derives a decryption key from it is plaintext with extra steps. Confidentiality needs a real app-layer key exchange (e.g. keys delivered out-of-band, or a proper ECDH scheme); if you cannot do that, say so in the UI rather than shipping theater.
6. **Inventing a fallback when derivation fails.** If `deriveTopicId` (or any kernel call) throws, surface the error. Substituting a locally-made-up id “so the UI keeps working” silently diverges that client from every other peer on the topic — the failure mode is invisible no-delivery (HARD RULE 1).

Verify with two clients — the single-client illusion

Do not declare an Axona app working from one client. **Every mistake in the list above renders perfectly on a single screen** — the app connects, the UI looks right, your own messages appear (your client shows you your own state, not the network’s). Distributed defects are only visible between two clients. The minimum gate before “done”, in order:

1. **Delivery.** Two clients (separate browsers or profiles), same topic: a message sent from A renders on B within seconds.
2. **Retraction propagates.** A kills its own message → it disappears on B via the `{ deleted: true }` marker — not merely locally on A.
3. **Share through your own surface.** A shares a topic through whatever your app uses (invite, directory ad, QR, link) and B joins through it. B must see **A’s content**. A clean-looking but empty topic on B means the shared descriptor was partial (mistake 3) — this is the test that catches it.
4. **Recency honesty.** After history has accumulated, a freshly-started client’s presence/“last seen” surface shows only currently-active parties — replayed history must not resurrect the departed (mistake 4).
5. **Persona switch is free.** Switching the active user/author must not reconnect the peer — watch the connection state while switching (mistake 2).
6. **Failure is loud.** Feed one malformed topic descriptor: a visible error and no subscription — never a silently invented topic (mistake 6).
7. **Both engines.** Run the gate in a Chromium browser AND in Firefox — real environment differences exist (see Troubleshooting).

Troubleshooting rules

- **Connected but nothing arrives** → publisher and subscriber derived different topic IDs. Diff the descriptors field-by-field (including an omitted vs named `region`), or → published before `ready()`.
- **Cannot mix BigInt and other types** → pre-4.14 kernel with a hex id passed to `NeuronNode`; on this version the hex id is accepted directly (and `connect()` avoids the question entirely).
- **Works on localhost, fails deployed** → page not HTTPS (`crypto.subtle` unavailable).
- **Firefox only: “ICE failed, your TURN server appears to be broken” repeating, peers stuck at ~1, no delivery** → the dev server is bound to IPv6 loopback (`::1`) — Vite’s default. Firefox gathers ZERO ICE candidates on a `::1`-origin page, so every mesh dial fails; Chromium is unaffected. Fix: `server: { host: '127.0.0.1' }` in `vite.config.js` (or serve on an IPv4 address). The TURN server is fine — do not debug it.
- **Large message silently missing for some receivers** → over the 15 KB reliable floor. Use `std/chunk`.
- **User identity resets every reload** → author not persisted. Use `createAuthorIdentity({ persistAs })`.
- **WS close 4426** → version mismatch; reinstall the pinned tag.

Nodes can refuse work (kernel 4.46.0+)

A node is allowed to say **no** to a role. If it is newly joined (still in its grace period), or genuinely at capacity, it declines and the role goes to someone who can serve it. Three things follow that matter when you are reading logs or reasoning about delivery:

- **Refusal is not loss.** A declined publish is forwarded to a node that can take it. Only if there is *no* such node does it stop — and then it is logged `undeliverable` rather than retried, so it shows up instead of hanging.
- **The floor overrides refusal when there is no alternative.** You will see `admitted-despite` in logs: the node took a role it did not want because dropping the data was the worse option. That is working as designed, not a bug.
- **Capacity is measured, not counted.** A node with hundreds of roles that services them on time is healthy. Judge nodes by `servicePressure` / `helloPressure` from `peer.health().admission.capacity`, never by role count.

One refusal is absolute: **a bridge never holds a topic role**, at any load. See the Services Guide.

Environment notes

- Browser: HTTPS only. Node: v20+; the same `webTransport` connects to the bridge over WSS (the WebRTC mesh is browser-side; Node peers converse via the bridge and routing).
- **Bundlers (Vite/webpack): alias `node-datachannel` and `node-datachannel/polyfill` to an empty stub module** (`export default {}`) for browser builds. It is the kernel's Node-only WebRTC dependency; the browser never executes that import path, but the bundler must still be able to resolve it.
- **After changing the kernel pin, clear the bundler's dependency cache** (Vite: delete `node_modules/.vite`) and restart the dev server. The stale-cache failure mode — the old kernel silently served under the new version number — has repeatedly cost real debugging time.
- The testnet bridge is `wss://testnet.axona.net` (kernel 4.48.0, wire 4.0) — the network this grounding targets. Production (`wss://bridge.axona.net`) runs the same wire-4 line, typically one release behind; the two are wire-compatible but SEPARATE networks (a peer joins one or the other).
- Multiple tabs = independent peers (fine, but each is a separate node).

End of grounding (tier 1). Need the parts of the API this file doesn't cover — persistence, snapshots, custom transports, full error taxonomy, metrics aggregation, the behavioral/timing model, MCP tools? Load tier 2: Axona-AI-Reference-v4.38.0.md, sectioned for selective loading. Human-oriented companions: the Axona Quick Start, Programmer Guide, and API Reference (same version).